OXFORD AQA

INTERNATIONAL QUALIFICATIONS

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I declare this is my own work.

INTERNATIONAL A-LEVEL
CHEMISTRY (9620)

Unit 4: Organic 2 and Physical 2

Wednesday 11 June 2025 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do **not** write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
TOTAL	

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ANSWER IN THE SPACES PROVIDED

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Answer **all** questions in the spaces provided.

0 1 A student reacts lumps of calcium carbonate with hydrochloric acid to determine the initial rate of reaction.

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

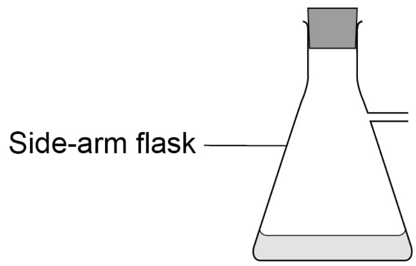
Method

- Add 45 cm^3 of 0.10 mol dm^{-3} hydrochloric acid to a side-arm flask.
- Weigh out about 3 g of calcium carbonate and add it to the flask.
- Put a stopper in the flask.
- Record the volume of gas formed every 10 seconds.

0 1 . 1 Complete the diagram in **Figure 1** to show how the volume of gas formed is measured.

[1 mark]

Figure 1



0 1 . 2 Suggest why the student does not weigh out the calcium carbonate more accurately.

[1 mark]

____

____

0 1 . 3 The student measures 45 cm^3 of hydrochloric acid using a 100 cm^3 measuring cylinder with an uncertainty of ± 0.5 cm^3

Calculate the percentage uncertainty in the measurement of the hydrochloric acid.

[1 mark]

Percentage uncertainty ____________________

Turn over ►

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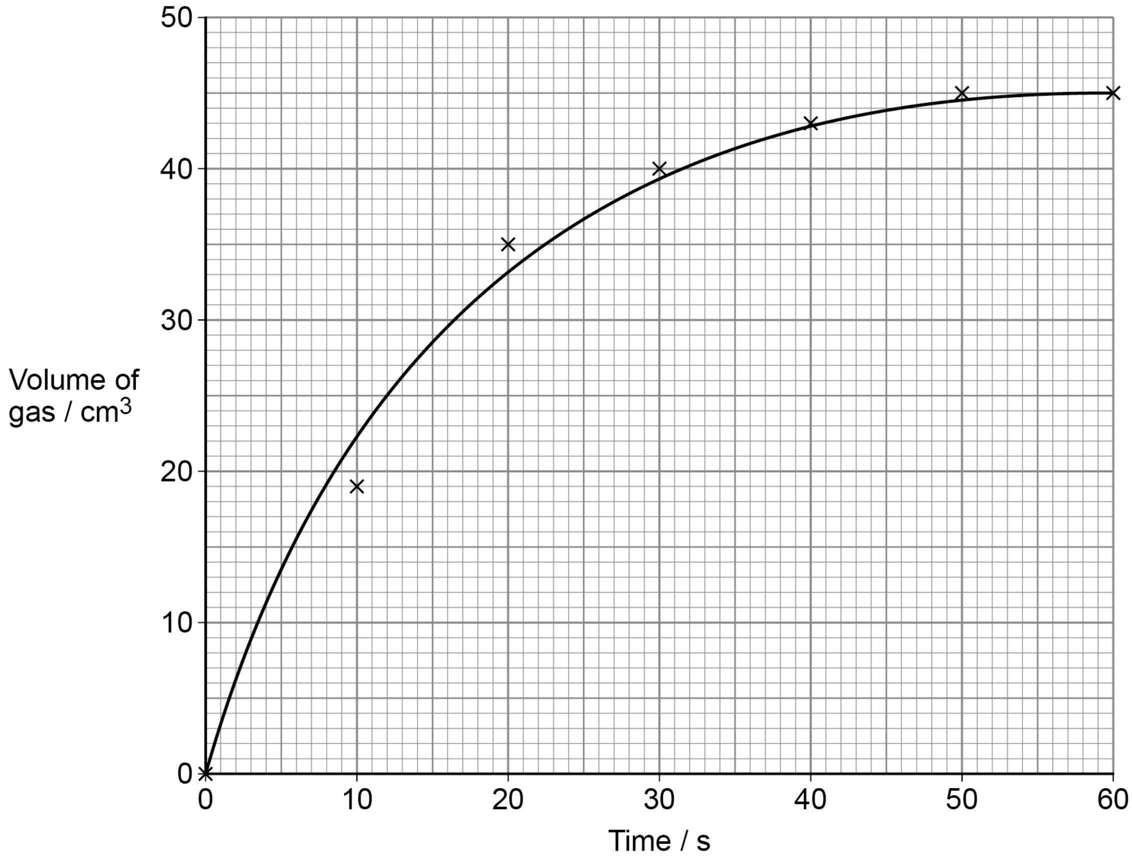
Figure 2 shows a graph of the student's results.

Figure 2



0 1 . 4 Draw a tangent on **Figure 2** and use it to determine the initial rate of reaction.

State the units.

[3 marks]

Initial rate ____________________________

Units ____________________________

Do not write outside the box

0 1 . 5 The total volume of carbon dioxide gas collected is less than expected due to some gas being lost.

Suggest what the student should do to reduce the volume of gas lost.

[1 mark]

0 1 . 6 Give a different method to determine the initial rate of this reaction.
You should **not** discuss gas collection in your answer.

[1 mark]

8

Turn over for the next question

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0 2 This question is about rates of reaction.

0 2 . 1 The initial rate of a reaction is measured in a series of experiments at constant temperature.

$$A + B \rightarrow 2C + D$$

The results are shown in **Table 1**.

Table 1

Experiment	**Initial [A] / mol dm^{-3}**	**Initial [B] / mol dm^{-3}**	**Initial rate / mol dm^{-3} s^{-1}**
1	0.20	0.15	1.2×10^{-2}
2	0.30	0.15	2.7×10^{-2}
3	0.40	0.30	4.8×10^{-2}
4	0.50	0.10	To be calculated

Use **Table 1** to deduce the order of reaction with respect to **A** and the order of reaction with respect to **B**.

Calculate the initial rate for **Experiment 4**.

[3 marks]

Order with respect to **A** ____________________

Order with respect to **B** ____________________

Initial rate for **Experiment 4** ____________________ mol dm^{-3} s^{-1}

Do not write outside the box

In a different reaction

$$E(g) + 3F(g) \rightarrow EF_3(g)$$

The reaction is first order with respect to **E** and second order with respect to **F**.

0 2 . 2 Give the rate equation for the reaction.

State the units of the rate constant, *k*

[2 marks]

Rate =

Units of the rate constant, *k* ____________________

0 2 . 3 The reaction takes place by a three-step mechanism.
Step 2 is the rate determining step.

Suggest **one** mechanism for the reaction.

[2 marks]

Step 1 ____________ → ____________

Step 2 ____________ → EF_2

Step 3 $EF_2 + F \rightarrow EF_3$

Question 2 continues on the next page

Do not write outside the box

0 2 . 4 The reaction is repeated at different temperatures and a value for the rate constant (k) is determined at each temperature.

A graph of ln k against $\frac{1}{T}$ is plotted.
A line of best fit for this graph has a gradient with the value of -2.50×10^4 K

Calculate the activation energy (E_a) in kJ mol^{-1}, for this reaction.

Use the equation $\ln k = \frac{-E_a}{RT} + \ln A$

The gas constant, $R = 8.31$ J K^{-1} mol^{-1}

[2 marks]

E_a ________________________ kJ mol^{-1}

9

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0 3 Hydrazine (N_2H_4) decomposes in the presence of a catalyst

$$N_2H_4(g) \rightleftharpoons N_2(g) + 2\,H_2(g)$$

0 3 . 1 Write an expression for the equilibrium constant (K_p) for the reaction.

Deduce the units of K_p when all the partial pressures are given in kPa

[2 marks]

K_p

Units ____________________________

0 3 . 2 3.0 mol of N_2H_4 are placed in a container with the catalyst and heated to a known temperature.

The equilibrium mixture contains 1.2 mol of H_2

Calculate the amount, in mol, of N_2H_4 and the amount, in mol, of N_2 in the equilibrium mixture.

[2 marks]

Amount of N_2H_4 ____________________________ mol

Amount of N_2 ____________________________ mol

Turn over ►

Do not write outside the box

A different equilibrium mixture is formed by heating some hydrazine in the presence of the catalyst to a different temperature.

$$N_2H_4(g) \rightleftharpoons N_2(g) + 2H_2(g)$$

This equilibrium mixture has a total pressure of 150 kPa

0 3 . 3 **Table 2** shows the mole fractions and partial pressures for the components of the gaseous mixture.

Complete **Table 2**.

[2 marks]

Table 2

	N_2H_4	N_2	H_2
Mole fraction	0.10		
Partial pressure / kPa			90

0 3 . 4 The pressure of the equilibrium mixture is increased.

State the effect of this increase on the position of equilibrium and on the value of K_p

[2 marks]

Effect on position of equilibrium ______________________________

Effect on K_p ______________________________

8

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0 4 This question is about carboxylic acids and their derivatives.

0 4 . 1 Complete **Table 3** to name the strongest type of intermolecular force that occurs between the pairs of molecules.

[2 marks]

Table 3

Pairs of molecules	Strongest type of intermolecular force
Ethyl ethanoate and ethanoyl chloride	
Ethanamide and ethanoic acid	

0 4 . 2 Which carboxylic acid derivative is the most reactive with water?

Tick (✓) **one** box.

[1 mark]

$(CH_3CH_2CO)_2O$ ☐

CH_3CH_2COOH ☐

CH_3CH_2COCl ☐

Question 4 continues on the next page

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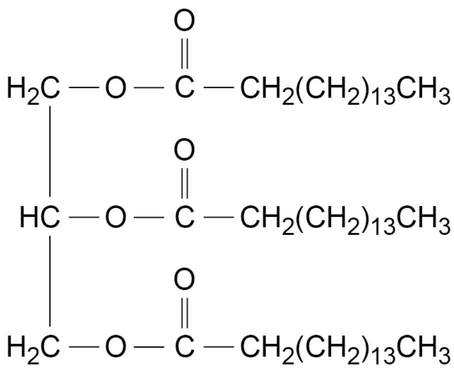
A triglyceride is shown.



0 4 . 3 The triglyceride reacts with methanol in alkaline conditions to form propane-1,2,3-triol and one other product.

Give the formula of this product **and** state a use for this type of product.

[2 marks]

Product ______________________________

Use of product ______________________________

0 4 . 4 The triglyceride reacts with sodium hydroxide to form propane-1,2,3-triol and one other product.

Give the formula of this product **and** state a use for this type of product.

[2 marks]

Product ______________________________

Use of product ______________________________

7

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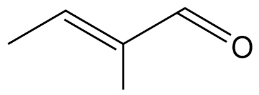
0 5 This question is about aldehydes and isomers.



Compound G

0 5 . 1 Select the full IUPAC name of **G** from the list.

Tick (✓) **one** box.

[1 mark]

Z-3-methylbut-2-enal ☐

Z-2-methylbut-2-enal ☐

E-2-methylbut-2-enal ☐

E-3-methylbut-2-enal ☐

0 5 . 2 Draw the structure of an isomer of **G** that decolourises bromine water **and** is

- a functional group isomer of **G** but **not** an alcohol
- **not** a chain isomer of **G**.

[1 mark]

Do not write outside the box

0 5 . 3 Explain why $NaBH_4$ can reduce the carbonyl group in **G** but can **not** reduce the alkene group in **G**.

[2 marks]

Ethanal can be converted into $CH_3CH(OH)COOH$ (lactic acid) by a two-stage process.

Stage 1 $CH_3CHO + HCN \rightarrow CH_3CH(OH)CN$
2-hydroxypropanenitrile

Stage 2 $CH_3CH(OH)CN + HCl + 2H_2O \rightarrow CH_3CH(OH)COOH + NH_4Cl$

M_r $CH_3CH(OH)COOH = 90.0$

0 5 . 4 Suggest **one** essential piece of safety equipment, other than safety goggles or gloves, that must be used in **Stage 1**.

[1 mark]

0 5 . 5 Calculate the percentage atom economy for the formation of $CH_3CH(OH)COOH$ in **Stage 2**.

[1 mark]

Percentage atom economy ____________________

Do not write outside the box

0 5 . 6 In **Stage 1**, 10.0 cm^3 of CH_3CHO gives a 71.5% yield of $CH_3CH(OH)CN$
In **Stage 2**, the mass of $CH_3CH(OH)COOH$ formed is 4.35 g
Density of CH_3CHO = 0.780 g cm^{-3}

Calculate the percentage yield for the formation of $CH_3CH(OH)COOH$ in **Stage 2**.

[5 marks]

Percentage yield ________________________

0 5 . 7 Draw **both** enantiomers of 2-hydroxypropanenitrile and show their relationship to each other.

[1 mark]

12

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0 6 This question is about amino acids.

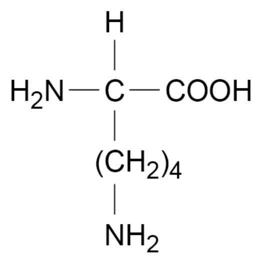
0 6 . 1 State what is meant by a zwitterion.

[1 mark]

0 6 . 2 State **one** physical property of amino acids at room temperature.

[1 mark]

0 6 . 3 The amino acid lysine has the following structure.

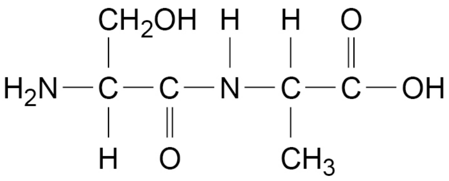


Draw the structure of lysine in a solution at pH = 12

[1 mark]

Do not write outside the box

0 6 . 4 A dipeptide is shown.

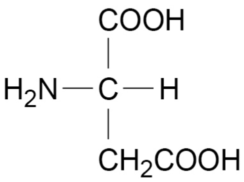


Give the structures of the **two** products formed after boiling this dipeptide in an **excess** of hydrochloric acid.

[2 marks]

0 6 . 5 Draw the skeletal formula of the organic product formed when aspartic acid reacts with an **excess** of methanol and a little concentrated sulfuric acid.

[1 mark]



Aspartic acid

6

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0 7 This question is about amines.
Ethane can be converted into ethylamine in two steps.

$$CH_3CH_3 \xrightarrow{\textbf{Reaction 1}} CH_3CH_2Br \xrightarrow{\textbf{Reaction 2}} CH_3CH_2NH_2$$

0 7 . 1 Give the reagent and **one** condition for **Reaction 1**.

State the type of intermediate involved in the mechanism for **Reaction 1**.

[3 marks]

Reagent ____________________

Condition ____________________

Intermediate ____________________

0 7 . 2 In **Reaction 2**, ethylamine is produced by the reaction of bromoethane with an excess of concentrated ammonia solution.

$$CH_3CH_2Br + 2NH_3 \rightarrow CH_3CH_2NH_2 + NH_4Br$$

This occurs in two steps.

Step 1

$$CH_3CH_2Br + NH_3 \rightarrow CH_3CH_2\overset{+}{N}H_3 + Br^-$$

Step 2

$$CH_3CH_2\overset{+}{N}H_3 + NH_3 \rightarrow CH_3CH_2NH_2 + NH_4^+$$

State **and** explain the role of ammonia in **Step 1** and the role of ammonia in **Step 2** of this reaction.

[4 marks]

Role of ammonia in **Step 1** ____________________

Explanation ____________________

____

____

Role of ammonia in **Step 2** ____________________

Explanation ____________________

____

____

Do not write outside the box

0 7 . 3 Under different conditions, ammonia can react with an **excess** of CH_3CH_2Br

Draw the structure of the final product.

[1 mark]

0 7 . 4 Ethylamine can also be formed from bromomethane in a different two-step process.

$$CH_3Br \xrightarrow{\textbf{Reaction 3}} \text{Compound } \mathbf{Q} \xrightarrow{\textbf{Reaction 4}} CH_3CH_2NH_2$$

Identify compound **Q**.

Give the reagent and a necessary condition for **Reaction 4**.

[2 marks]

Compound **Q** ____________________

Reagent and condition for **Reaction 4** ____________________

0 7 . 5 State why the ethylamine made in Question **07.4** is purer than the ethylamine made in Question **07.2**.

[1 mark]

____

____

____

11

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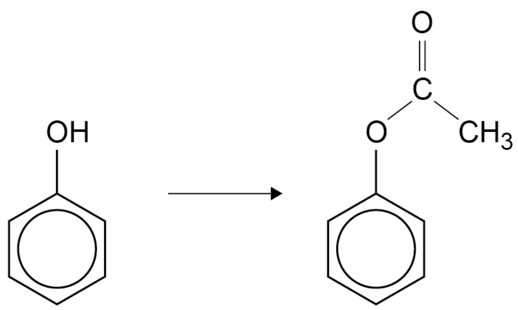
0 8 Phenol can react with ethanoyl chloride to make phenyl ethanoate.

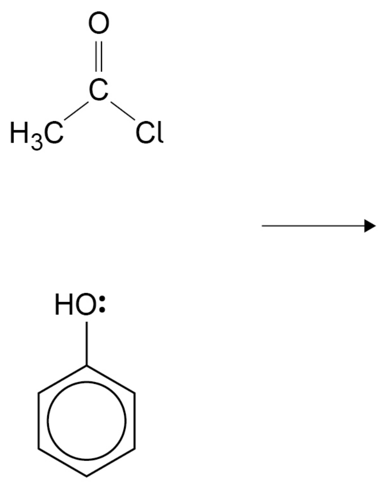


0 8 . 1 Name **and** complete the mechanism for this reaction.

[4 marks]

Mechanism name ______________________________

Mechanism



0 8 . 2 Give **one** reason why ethanoic anhydride is used instead of ethanoyl chloride in industry for this reaction.

[1 mark]

____

Do not write outside the box

0 8 . 3 Phenol and ethanoyl chloride, in the presence of an $AlCl_3$ catalyst, react to form a mixture of three isomers.
The reaction involves an initial step when $AlCl_3$ reacts with ethanoyl chloride to form a reactive species.
This reactive species then reacts with the benzene ring in phenol.

Write an equation for the formation of the reactive species.

Name **and** outline the mechanism for the reaction of the reactive species with the benzene ring in phenol to form one of the isomers.

[5 marks]

Equation

____

Name of mechanism ______________________________

Mechanism

10

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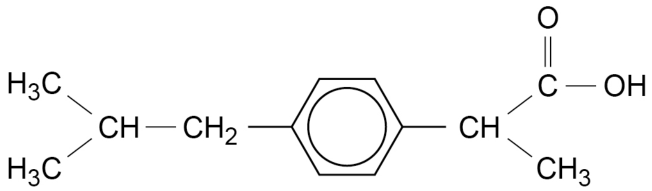
0 9 **Compound S** can be analysed using spectroscopy.

Compound S



0 9 . 1 Write an asterisk (*) on the structure above to identify any chiral carbon atom(s) in **S**.

[1 mark]

0 9 . 2 One functional group in **S** reacts with $NaHCO_3$ to form carbon dioxide.

Give the **two** wavenumber ranges of the absorbances in the infrared spectrum of **S** that confirm the presence of this functional group.
Use **Table A** on the Chemistry Data Sheet.

[1 mark]

____

____

0 9 . 3 How many peaks appear in the ^{13}C NMR spectrum of **S**?

Tick (✓) **one** box to select the correct answer.

[1 mark]

7 ☐ **10** ☐ **13** ☐

Do not write outside the box

0 9 . 4 Identify the compound that is used as the standard when recording a ^{1}H NMR spectrum.

The chemical shift of the standard used in ^{1}H NMR spectroscopy is zero by definition.

State **one** other reason why this standard is used in ^{1}H NMR.

[2 marks]

Standard ______________________________

Reason ______________________________

____

Question 9 continues on the next page

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0 9 . 5 The structure of **Compound S** is repeated below.

Compound S

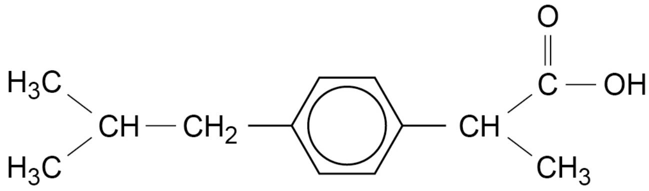
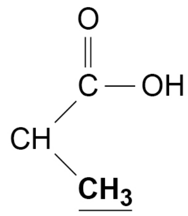
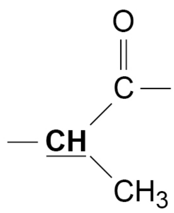


Table 4 shows data for four of the peaks in the ^{1}H NMR spectrum of **Compound S**. None of these peaks are caused by the hydrogen atoms on the benzene ring.

Complete **Table 4** to show the splitting pattern, integration value and section of the molecule producing the peak.
Underline the relevant hydrogen(s) in the section of the molecule.

[4 marks]

Table 4

Splitting pattern	Integration value	Section of the molecule producing the peak
	6	
		
	2	
		

9

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